

Mohamed Rayen Fekih

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Senior AI Engineer | 5+ Years of Gen AI and Data Science | +216 52591307

Skills

Languages: Python, SQL, Spark

Frameworks: LangChain, Pandas, Pyspark, Numpy, Scikit-learn, LangChain

Platforms/Tools: Google Cloud Platform, Vertex AI, n8n, Docker, CI/CD, GIT, Tableau, Gitlab

Soft Skills: Trilingual Communicator (**Arabic, English, French**)

Experience

AI Engineer, iHorizons – Qatar, (Remote, Full time)

July 2025– Present

Fatwa Semantic Search

- Developed a Fatwa semantic search engine for the **IslamWeb FatwaTok** app.
- Processed and indexed **165k** plus fatwas using Gemini batch inference and BigQuery-based vector search.
- Orchestrated a **Gemini 3 batch inference pipeline** to summarize 165k fatwas using **Vertex AI**.
- Designed a scalable **Text-to-Speech** pipeline generating Arabic audio at scale while preserving Quran Tajweed rules.
- Developed a Quran detector to process the Quran verses according to the strict religious rules.
- Deployed the full production pipeline on Google Cloud Platform using **Cloud Run, BigQuery, Cloud Storage, Vertex AI, and Cloud Scheduler**.

IslamWeb RAG Chatbots

- Developed an **Arabic RAG chatbot** that specializes in Arabic religious rulings (Fatwas) for islamweb.com.
- Implemented a **Human-in-the-loop** framework by implementing an intent detection **Agent** that escalates the conversation to a scholar (Mufti) when needed.
- Generated embeddings using Gemini models and used BigQuery as a Vector Search.
- Integrated chatbots with **WhatsApp** and **Telegram**.

Khateeb Assistant - Sermon (Khutbah) Generation

- Built a multi-agent system for the **Qatar Ministry of Islamic Affairs** that generates Friday Sermons using **n8n**.
- Processed the Quran verses, Hadith, and Arabic Islamic books and embedded them with **Google RAG Engine** to use as a sermon reference.
- Implemented requirement extractor, Sermon generation, moderator, and summarizer LLM **Agents** that collaborate to produce the Friday's Sermon respecting the sensitive religious rules.
- Developed an **LLM-as-judge** framework to evaluate the Sermon and ensure it meets the Islamic rulings and the Khateeb's requirements.

Machine Learning Engineer, Infor – New York, USA (Remote, Full time)

May 2023– June 2025

GenAI Assistant Agent Using RAG and LLMs

- Developed an intelligent agentic system using **LangGraph** Studio for graph-based agents and integrated advanced function calling and dynamic response capabilities.
- Implemented a **memory management** system to store and retrieve relevant contextual information, enhancing the agent's long-term coherence and adaptability.
- Deployed the application using **Docker**, ensuring portability, scalability, and simplified management across environments.

Data Scientist, Cognira – Tunisia (On site, Full time)

Feb 2021 – May 2023

Retail Demand Forecasting

- Developed and deployed a retail **demand time series forecasting** engine for a leading retailer.
- Collaborated with clients to address technical challenges and deliver tailored, project-based solutions.
- Processed and cleaned dataset of 10 million+ transactions using **PySpark** and **SQL**.

Education

National Engineering School of Tunis (ENIT), MS in Computer Science

2016 - 2021